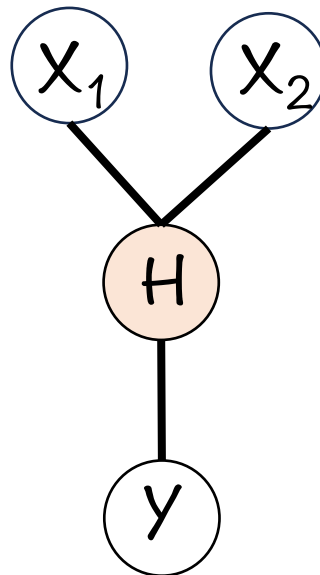
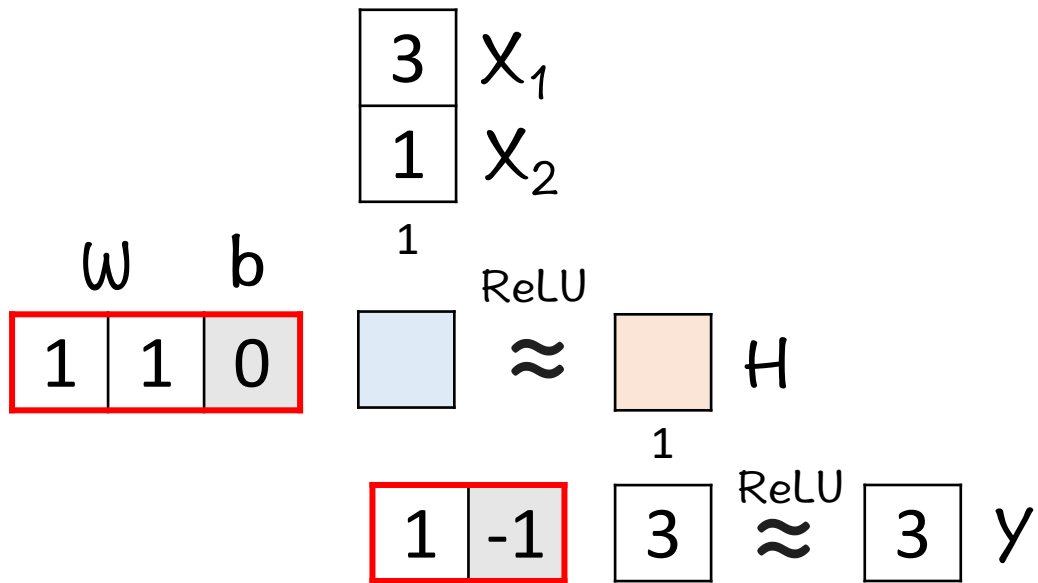


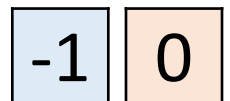
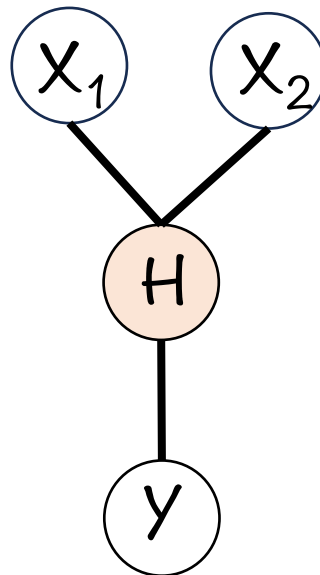
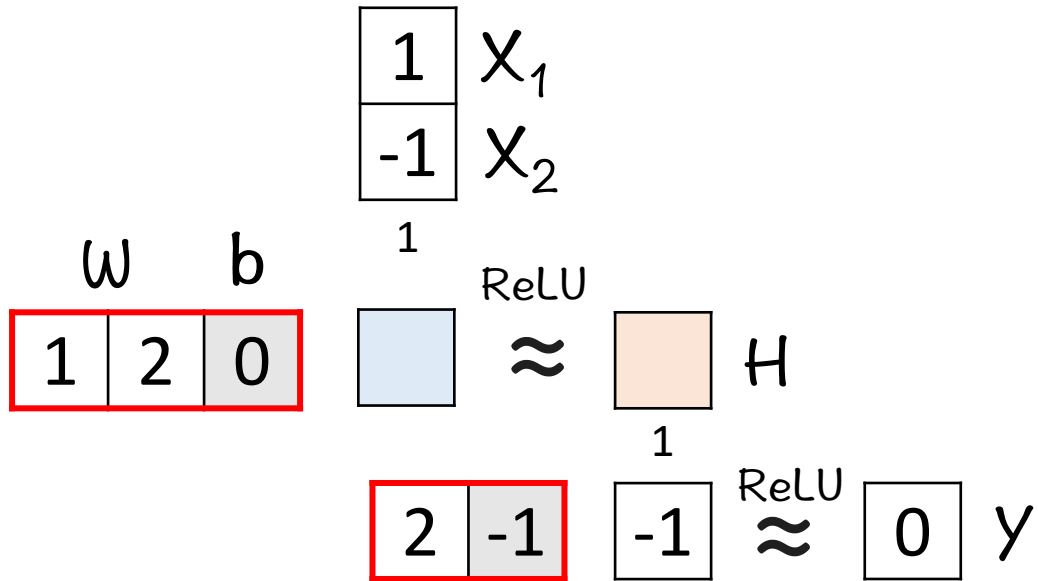
Hidden Layer

Exercise 1



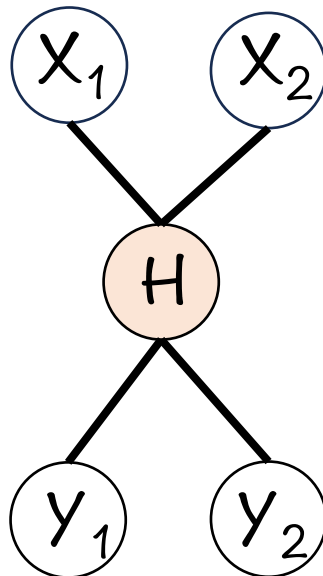
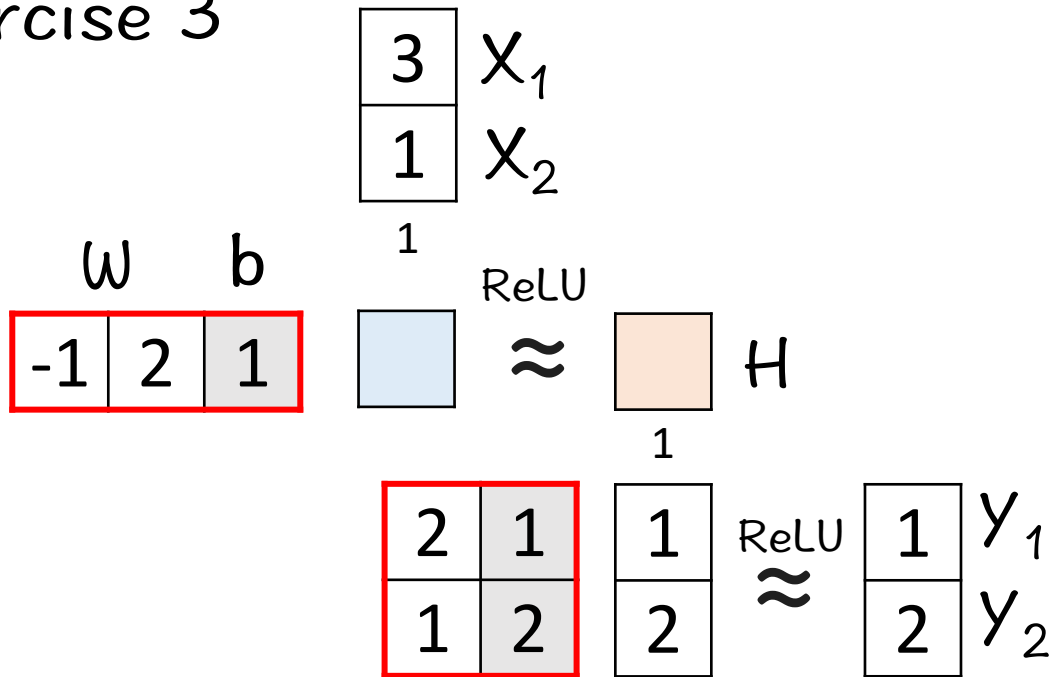
Hidden Layer

Exercise 2



Hidden Layer

Exercise 3

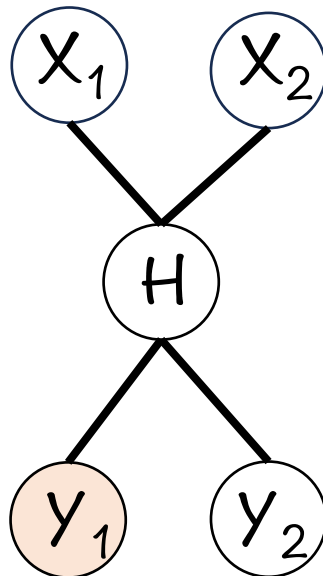


$\begin{bmatrix} 0 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix}$

Hidden Layer

Exercise 4

$$\begin{array}{c}
 \begin{array}{cc} w & b \end{array} \\
 \boxed{\begin{array}{|c|c|c|} \hline 1 & -1 & 0 \\ \hline \end{array}}
 \end{array}
 \begin{array}{c}
 \begin{array}{|c|} \hline 3 \\ \hline \end{array} X_1 \\
 \begin{array}{|c|} \hline 1 \\ \hline \end{array} X_2 \\
 1 \\
 \begin{array}{|c|} \hline 2 \\ \hline \end{array} \xrightarrow{\text{ReLU}} \begin{array}{|c|} \hline 2 \\ \hline \end{array} H \\
 1 \\
 \begin{array}{|c|c|} \hline 1 & -1 \\ \hline -1 & 1 \\ \hline \end{array} \xrightarrow{\text{ReLU}} \begin{array}{|c|} \hline -1 \\ \hline \end{array} \approx \begin{array}{|c|} \hline 0 \\ \hline \end{array} \begin{array}{l} y_1 \\ y_2 \end{array}
 \end{array}$$



$$\begin{array}{|c|} \hline 1 \\ \hline \end{array}
 \begin{array}{|c|} \hline 1 \\ \hline \end{array}$$

Hidden Layer

Exercise 5

$$\begin{array}{|c|} \hline 2 \\ \hline 3 \\ \hline \end{array} \begin{array}{l} x_1 \\ x_2 \end{array}$$

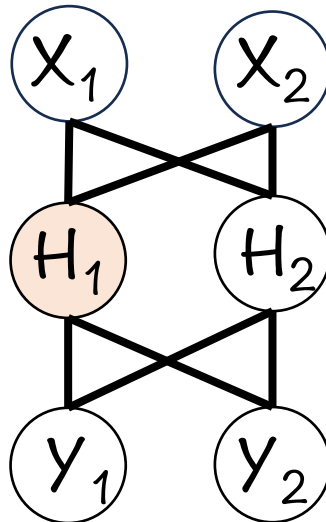
1

2	-1	1
1	1	-2

$$\begin{array}{|c|} \hline \text{ReLU} \\ \hline 3 \\ \hline \end{array} \approx \begin{array}{|c|} \hline H_1 \\ \hline 3 \\ \hline \end{array} \begin{array}{l} H_1 \\ H_2 \end{array}$$

1

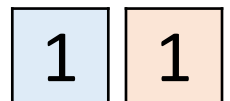
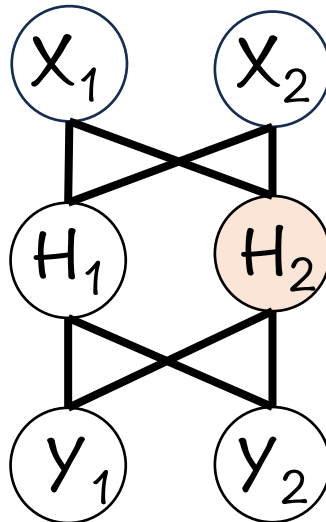
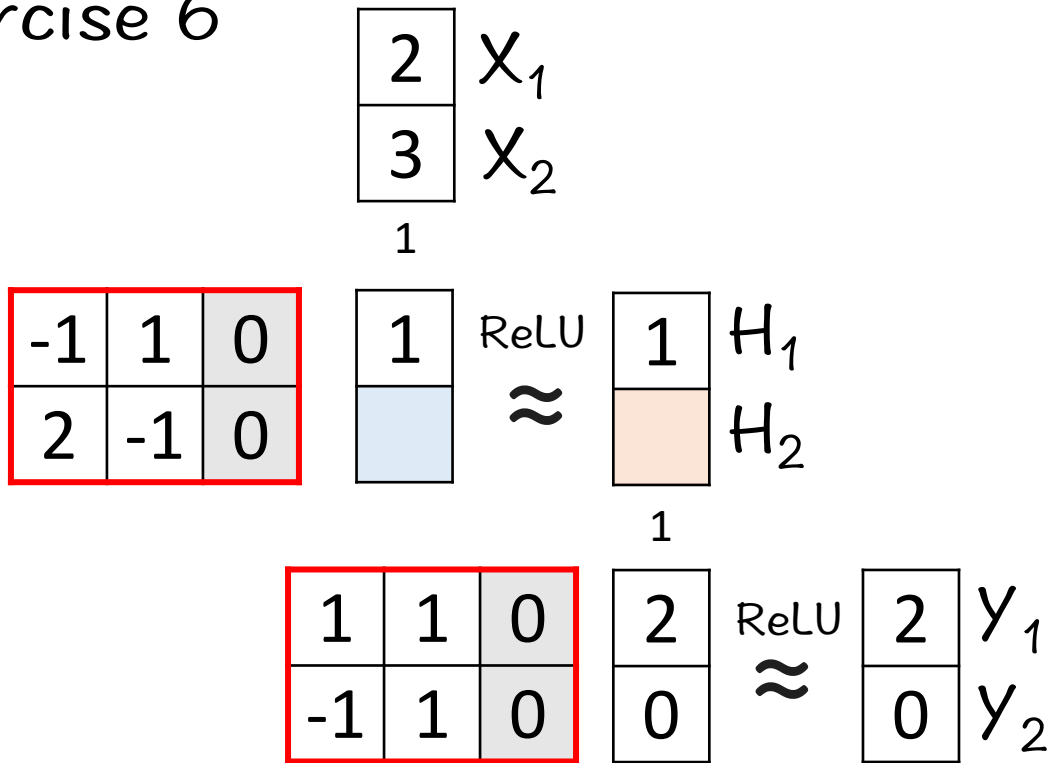
$$\begin{array}{|c|c|c|} \hline 1 & 1 & 0 \\ \hline 0 & 1 & -1 \\ \hline \end{array} \begin{array}{|c|} \hline 5 \\ \hline 2 \\ \hline \end{array} \begin{array}{l} \text{ReLU} \\ \approx \end{array} \begin{array}{|c|} \hline 5 \\ \hline 2 \\ \hline \end{array} \begin{array}{l} y_1 \\ y_2 \end{array}$$



$$\begin{array}{|c|} \hline 2 \\ \hline \end{array} \begin{array}{|c|} \hline 2 \\ \hline \end{array}$$

Hidden Layer

Exercise 6



Hidden Layer

Exercise 7

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$$

1

0	2	1
1	1	-1
1	0	-1

$$\begin{bmatrix} \\ 3 \\ 2 \end{bmatrix}$$

ReLU
 $\approx$

$$\begin{bmatrix} \\ 3 \\ 2 \end{bmatrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \end{matrix}$$

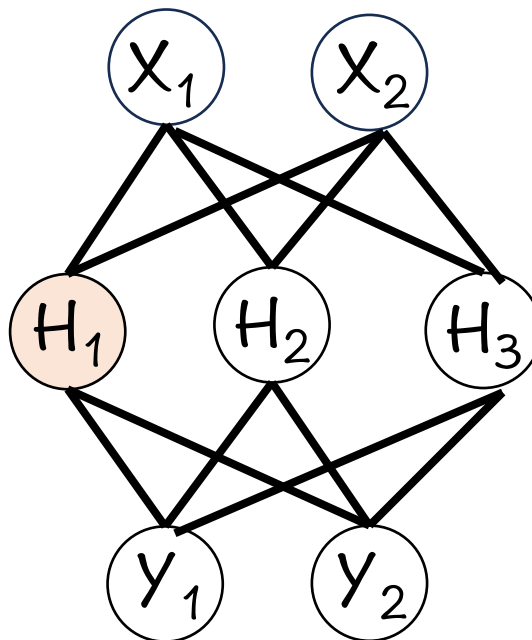
1

1	1	0	0
0	1	1	-1

$$\begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

ReLU
 $\approx$

$$\begin{bmatrix} 6 \\ 4 \end{bmatrix} \begin{matrix} Y_1 \\ Y_2 \end{matrix}$$



$$\begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Hidden Layer

Exercise 8

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$$

1

2	0	1
-1	1	0
1	1	-1

$$\begin{bmatrix} 7 \\ \text{ } \\ 3 \end{bmatrix}$$

ReLU
 $\approx$

$$\begin{bmatrix} 7 \\ \text{ } \\ 3 \end{bmatrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \end{matrix}$$

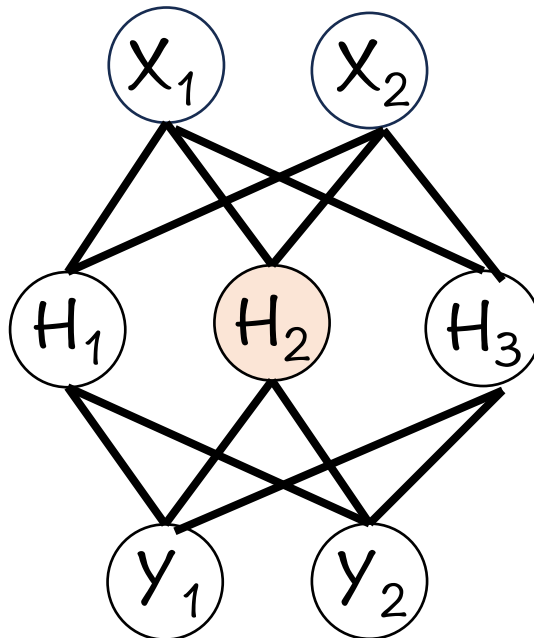
1

1	1	0	0
0	1	1	-1

$$\begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

ReLU
 $\approx$

$$\begin{bmatrix} 7 \\ 2 \end{bmatrix} \begin{matrix} Y_1 \\ Y_2 \end{matrix}$$



$$\begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

Hidden Layer

Exercise 9

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$$

1

2	0	1
-1	1	0
1	1	-1

$$\begin{bmatrix} 5 \\ -1 \\ \text{ } \end{bmatrix}$$

ReLU
 $\approx$

$$\begin{bmatrix} 5 \\ 0 \\ \text{ } \end{bmatrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \end{matrix}$$

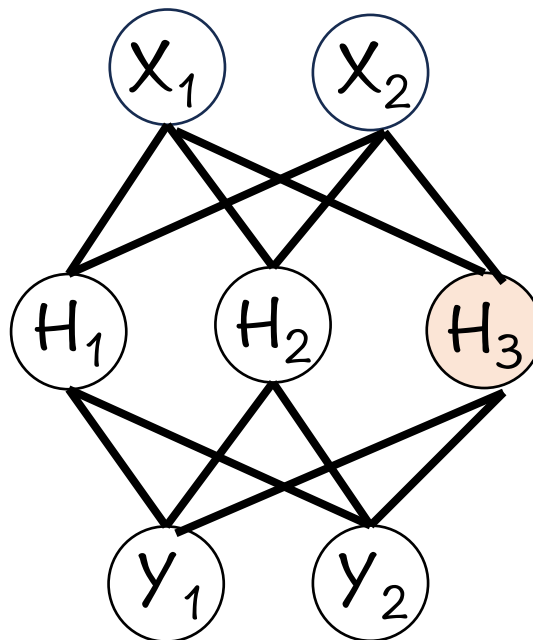
1

1	0	-1	0
0	1	1	-1

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

ReLU
 $\approx$

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \begin{matrix} Y_1 \\ Y_2 \end{matrix}$$



$$\begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Hidden Layer

Exercise 10

$$\begin{array}{|c|} \hline 2 \\ \hline 1 \\ \hline \end{array} \begin{array}{l} X_1 \\ X_2 \end{array}$$

1

2	0	1
-1	1	0
1	1	0

$$\begin{array}{|c|} \hline 5 \\ \hline -1 \\ \hline 3 \\ \hline \end{array}$$

ReLU
 $\approx$

$$\begin{array}{|c|} \hline 5 \\ \hline 0 \\ \hline 3 \\ \hline \end{array} \begin{array}{l} H_1 \\ H_2 \\ H_3 \end{array}$$

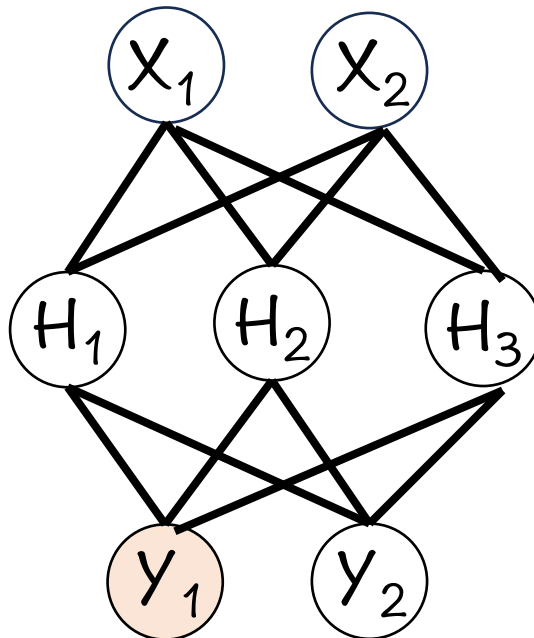
1

1	0	-1	-1
0	1	1	-1

$$\begin{array}{|c|} \hline \text{blue} \\ \hline 1 \\ \hline \end{array}$$

ReLU
 $\approx$

$$\begin{array}{|c|} \hline \text{orange} \\ \hline 1 \\ \hline \end{array} \begin{array}{l} Y_1 \\ Y_2 \end{array}$$



$$\begin{array}{|c|} \hline 1 \\ \hline \end{array} \begin{array}{|c|} \hline 1 \\ \hline \end{array}$$

Hidden Layer

Exercise 11

$$\begin{array}{|c|} \hline 3 \\ \hline 1 \\ \hline 2 \\ \hline \end{array} \begin{array}{l} X_1 \\ X_2 \\ X_3 \end{array}$$

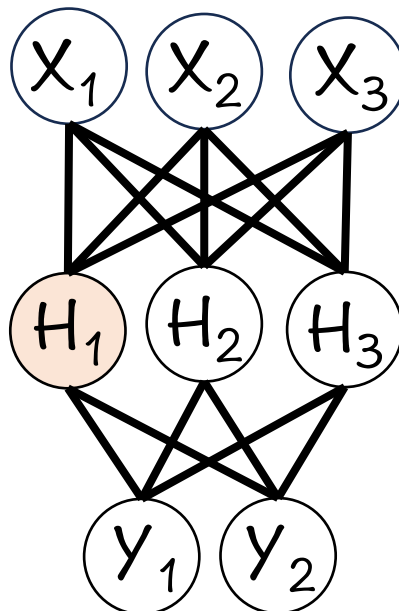
1

0	1	1	-1
0	1	0	-2
1	0	2	-1

$$\begin{array}{|c|} \hline \text{ReLU} \\ \hline -1 \\ \hline 6 \\ \hline \end{array} \approx \begin{array}{|c|} \hline H_1 \\ \hline 0 \\ \hline 6 \\ \hline \end{array} \begin{array}{l} H_1 \\ H_2 \\ H_3 \end{array}$$

1

$$\begin{array}{|c|c|c|c|} \hline 1 & 0 & 1 & 0 \\ \hline 0 & 1 & -1 & 0 \\ \hline \end{array} \begin{array}{|c|} \hline 8 \\ \hline -6 \\ \hline \end{array} \begin{array}{l} \text{ReLU} \\ \approx \end{array} \begin{array}{|c|} \hline 8 \\ \hline 0 \\ \hline \end{array} \begin{array}{l} Y_1 \\ Y_2 \end{array}$$



$$\begin{array}{|c|} \hline 2 \\ \hline \end{array} \begin{array}{|c|} \hline 2 \\ \hline \end{array}$$

Hidden Layer

Exercise 12

$$\begin{matrix} \boxed{2} & X_1 \\ \boxed{1} & X_2 \\ \boxed{-1} & X_3 \end{matrix}$$

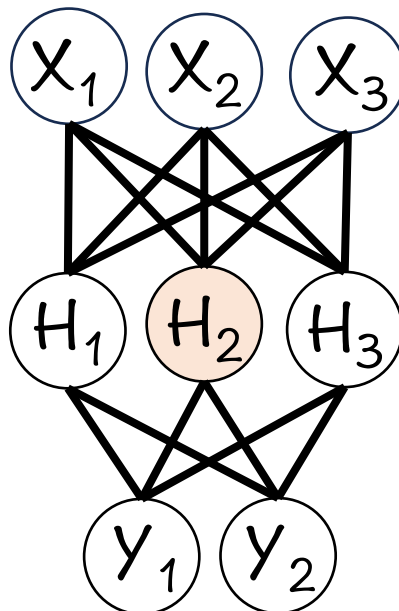
1

0	1	1	-1
-1	-1	0	0
1	0	0	-1

$$\begin{matrix} \boxed{-1} \\ \boxed{} \\ \boxed{1} \end{matrix} \text{ReLU} \approx \begin{matrix} \boxed{0} & H_1 \\ \boxed{} & H_2 \\ \boxed{1} & H_3 \end{matrix}$$

1

$$\begin{matrix} \boxed{1} & \boxed{2} & \boxed{1} & \boxed{1} \\ \boxed{0} & \boxed{1} & \boxed{0} & \boxed{1} \end{matrix} \text{ReLU} \approx \begin{matrix} \boxed{2} & Y_1 \\ \boxed{1} & Y_2 \end{matrix}$$



$$\begin{matrix} \boxed{-3} & \boxed{0} \end{matrix}$$

Hidden Layer

Exercise 13

$$\begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$$

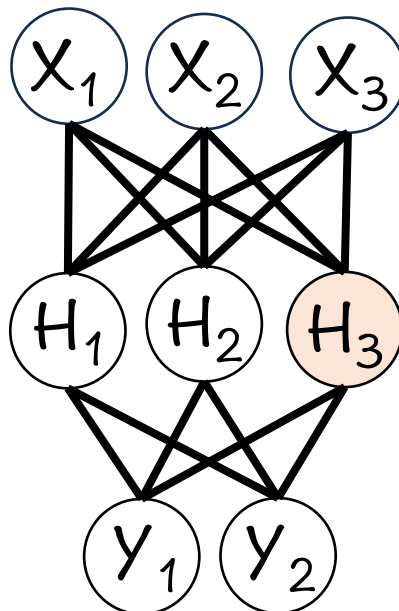
1

0	1	0	-1
1	1	1	0
0	-1	1	0

$$\begin{matrix} 1 \\ 6 \\ \end{matrix} \begin{matrix} \text{ReLU} \\ \approx \end{matrix} \begin{matrix} 1 \\ 6 \\ \end{matrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \end{matrix}$$

1

$$\begin{matrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{matrix} \begin{matrix} \text{ReLU} \\ \approx \end{matrix} \begin{matrix} 7 \\ 5 \end{matrix} \begin{matrix} Y_1 \\ Y_2 \end{matrix}$$



$$\begin{matrix} 1 & 1 \end{matrix}$$

Hidden Layer

Exercise 14

$$\begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$$

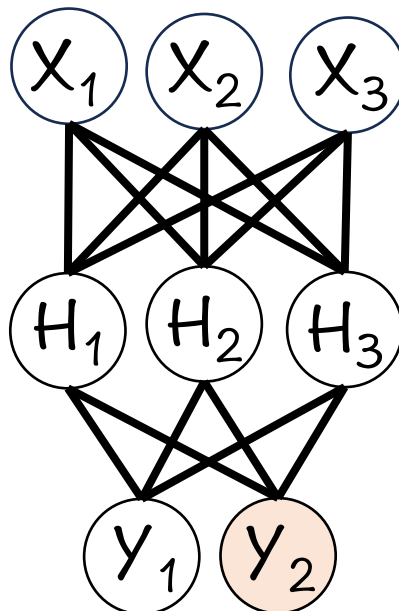
1

0	1	0	-1
1	1	1	0
-1	0	1	0

$$\begin{matrix} 1 \\ 6 \\ 2 \end{matrix} \begin{matrix} \\ \text{ReLU} \\ \approx \end{matrix} \begin{matrix} 1 \\ 6 \\ 2 \end{matrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \end{matrix}$$

1

$$\begin{matrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{matrix} \begin{matrix} 3 \\ \text{ReLU} \\ \approx \end{matrix} \begin{matrix} 3 \\ \text{ } \end{matrix} \begin{matrix} y_1 \\ y_2 \end{matrix}$$



$$\begin{matrix} 4 & 4 \end{matrix}$$

Hidden Layer

Exercise 15

$$\begin{matrix} 1 \\ 0 \\ 2 \end{matrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$$

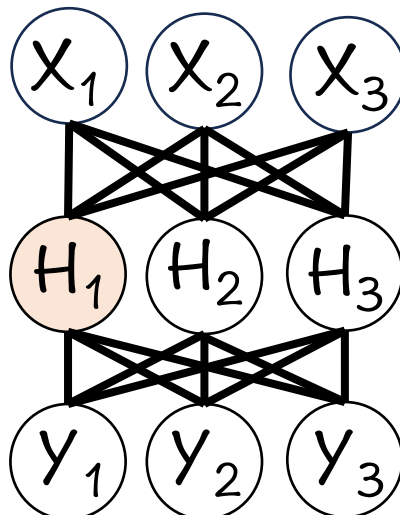
1

0	1	-1	-1
1	0	1	0
1	1	1	-1

$$\begin{matrix} \text{ReLU} \\ 3 \\ 2 \end{matrix} \approx \begin{matrix} H_1 \\ 3 \\ 2 \end{matrix} \begin{matrix} H_2 \\ H_3 \end{matrix}$$

1

$$\begin{matrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 1 & 1 & 1 & -1 \end{matrix} \begin{matrix} 3 \\ -2 \\ 4 \end{matrix} \xrightarrow{\text{ReLU}} \begin{matrix} 3 \\ 0 \\ 4 \end{matrix} \begin{matrix} Y_1 \\ Y_2 \\ Y_3 \end{matrix}$$



$$\begin{matrix} -3 \\ 0 \end{matrix}$$

Hidden Layer

Exercise 16

$$\begin{matrix} 1 \\ 0 \\ 2 \end{matrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$$

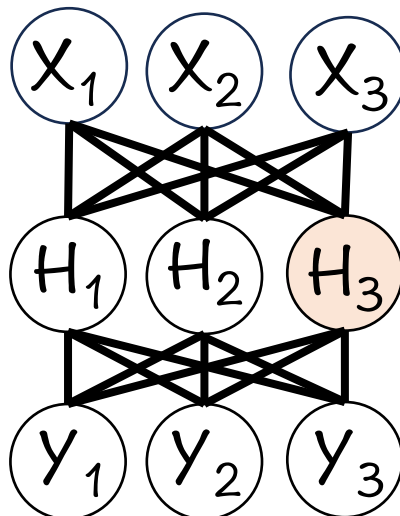
1

0	1	1	-1
1	0	1	0
0	1	0	-1

$$\begin{matrix} 1 \\ 3 \\ \end{matrix} \begin{matrix} \text{ReLU} \\ \approx \end{matrix} \begin{matrix} 1 \\ 3 \\ \end{matrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \end{matrix}$$

1

$$\begin{matrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 1 & 1 & 1 & -1 \end{matrix} \begin{matrix} \text{ReLU} \\ \approx \end{matrix} \begin{matrix} 4 \\ -2 \\ 3 \end{matrix} \begin{matrix} Y_1 \\ Y_2 \\ Y_3 \end{matrix}$$



-1	0
----	---

Hidden Layer

Exercise 17

$$\begin{matrix} \boxed{2} & X_1 \\ \boxed{1} & X_2 \\ \boxed{3} & X_3 \end{matrix}$$

1

0	1	1	0
1	1	0	0
1	0	1	0
1	0	0	-1

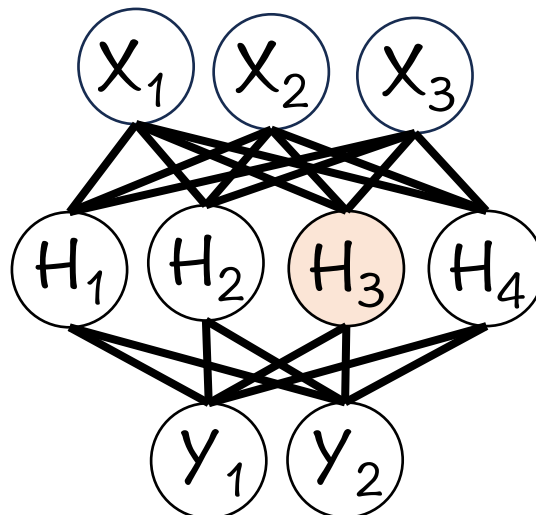
$$\begin{matrix} \boxed{4} \\ \boxed{3} \\ \boxed{} \\ \boxed{1} \end{matrix} \quad \text{ReLU} \approx$$

$$\begin{matrix} \boxed{4} & H_1 \\ \boxed{3} & H_2 \\ \boxed{} & H_3 \\ \boxed{1} & H_4 \end{matrix}$$

1

1	1	0	0	0
0	0	1	1	-1

$$\begin{matrix} \boxed{7} & \text{ReLU} & \boxed{7} & Y_1 \\ \boxed{5} & \approx & \boxed{5} & Y_2 \end{matrix}$$



$$\boxed{5} \quad \boxed{5}$$

Hidden Layer

Exercise 18

$$\begin{matrix} 2 \\ 1 \\ 3 \end{matrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$$

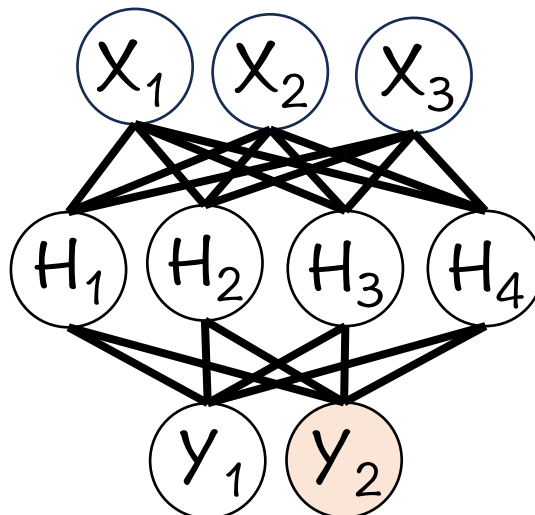
1

0	1	1	0
1	1	0	0
1	0	1	1
1	0	0	-1

$$\begin{matrix} 4 \\ 3 \\ 6 \\ 1 \end{matrix} \begin{matrix} \\ \\ \\ \end{matrix} \text{ReLU} \approx \begin{matrix} 4 \\ 3 \\ 6 \\ 1 \end{matrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \\ H_4 \end{matrix}$$

1

$$\begin{matrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & -2 \end{matrix} \begin{matrix} \\ \\ \\ \\ \end{matrix} \text{ReLU} \approx \begin{matrix} 7 \\ \end{matrix} \begin{matrix} Y_1 \\ Y_2 \end{matrix}$$



3	3
---	---

Hidden Layer

Exercise 19

$$\begin{matrix} 1 \\ -2 \\ 3 \end{matrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$$

1

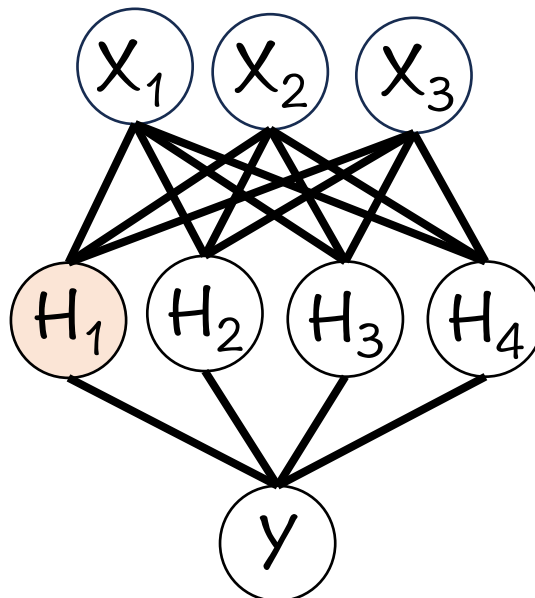
0	1	1	0
1	0	1	0
0	0	1	2
1	1	0	0

$$\begin{matrix} \text{ } \\ 4 \\ 5 \\ -1 \end{matrix} \text{ReLU} \approx$$

$$\begin{matrix} H_1 \\ 4 \\ 5 \\ 0 \end{matrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \\ H_4 \end{matrix}$$

1

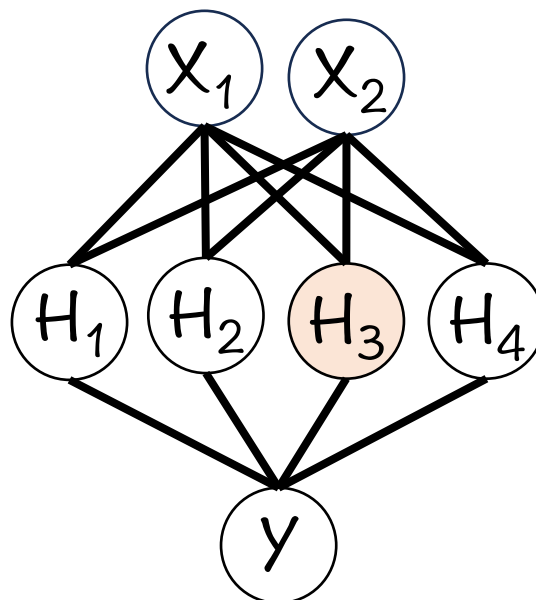
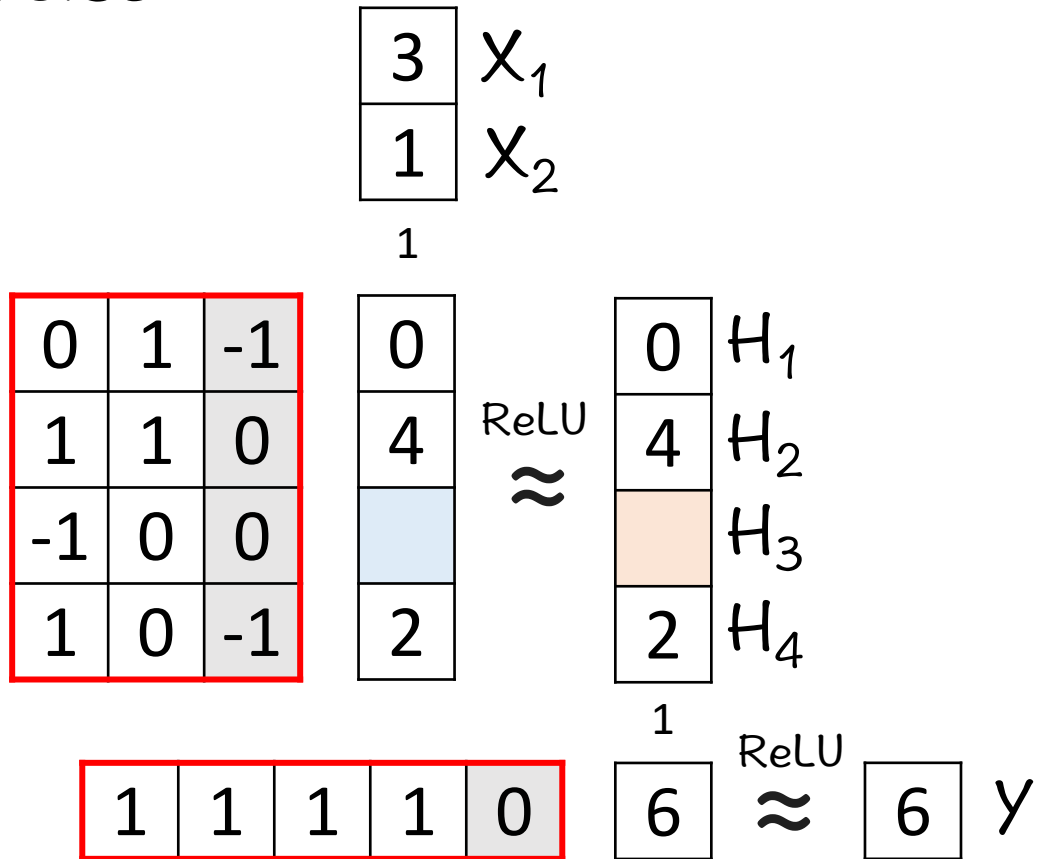
$$\begin{matrix} 1 & 1 & 1 & 0 & 0 \end{matrix} \text{ReLU} \approx \begin{matrix} 10 \end{matrix} \approx \begin{matrix} 10 \end{matrix} y$$



$$\begin{matrix} 1 & 1 \end{matrix}$$

Hidden Layer

Exercise 20

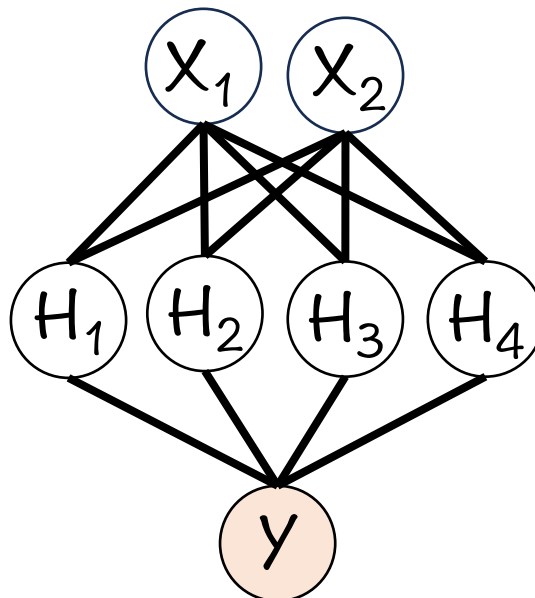
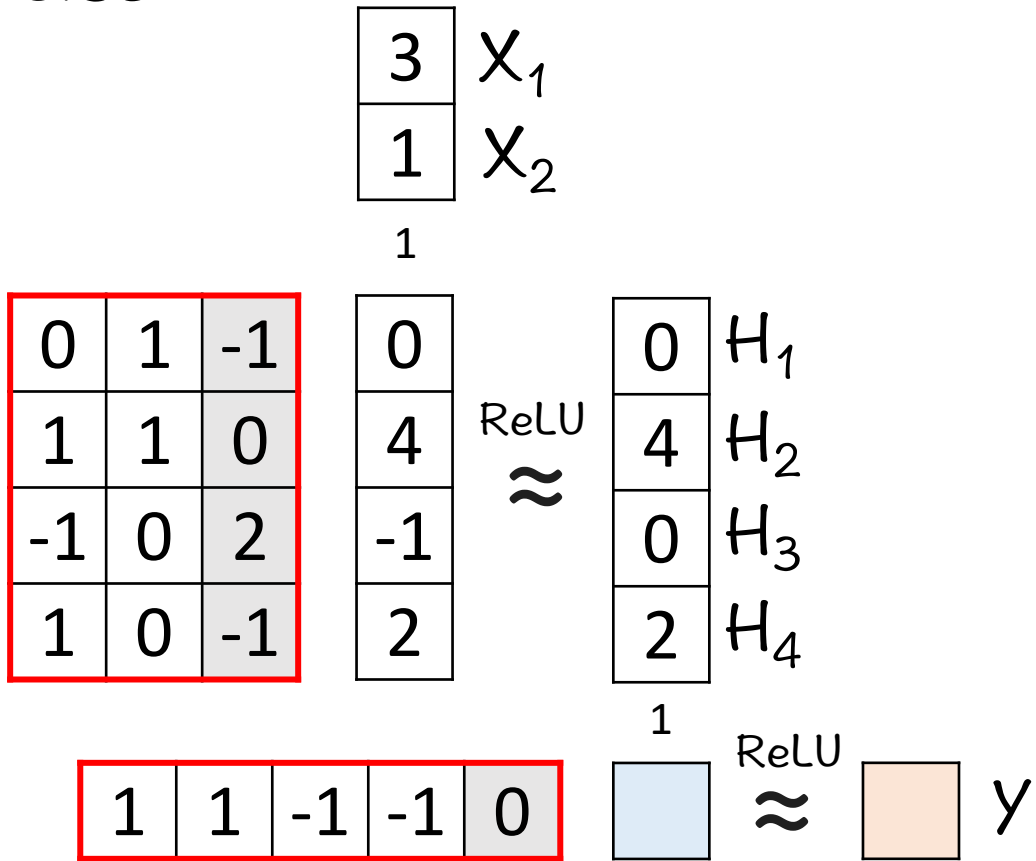


-3

0

Hidden Layer

Exercise 21



2

2

Hidden Layer

Exercise 22

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$$

1

0	0	-1
1	0	1
0	1	1
-1	0	3
1	1	2

$$\begin{bmatrix} -1 \\ 3 \\ 0 \\ 1 \\ \text{ } \end{bmatrix}$$

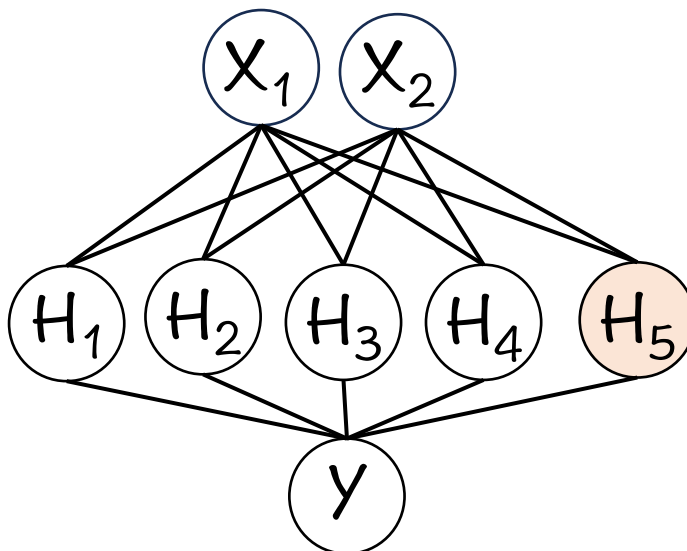
ReLU
 $\approx$

$$\begin{bmatrix} 0 \\ 3 \\ 0 \\ 1 \\ \text{ } \end{bmatrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \\ H_4 \\ H_5 \end{matrix}$$

1

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 7 \end{bmatrix} \xrightarrow{\text{ReLU}} \begin{bmatrix} 7 \end{bmatrix} y$$



$$\begin{bmatrix} 3 \end{bmatrix} \begin{bmatrix} 3 \end{bmatrix}$$

Hidden Layer

Exercise 23

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$$

1

0	0	-1
1	0	1
0	1	1
-1	0	3
1	1	0

-1
3
0
1
1

ReLU
 $\approx$

0	H_1
3	H_2
0	H_3
1	H_4
1	H_5

1

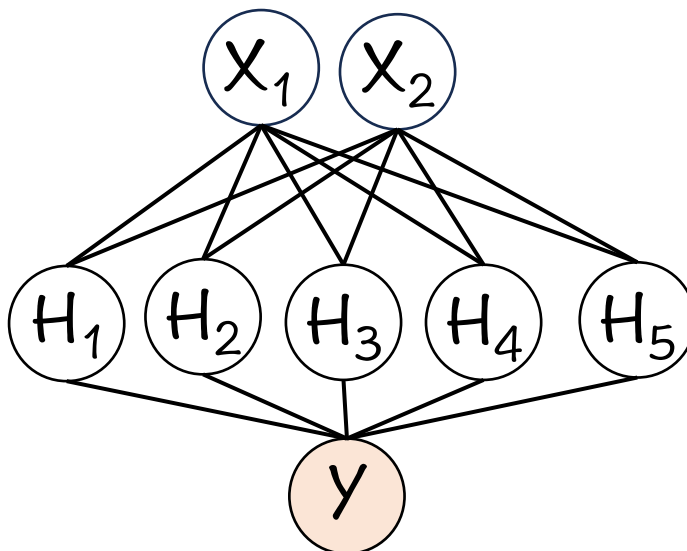
1	1	1	-1	-1	1
---	---	---	----	----	---

--

ReLU
 $\approx$

--

y



2	2
---	---

Hidden Layer

Exercise 24

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$$

1

0	1	0
1	0	0
-1	1	0
0	0	2
0	-1	1
1	-1	0

$$\begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \\ \text{ } \\ 1 \end{bmatrix}$$

ReLU
 $\approx$

$$\begin{bmatrix} 2 \\ 3 \\ 0 \\ 2 \\ \text{ } \\ 1 \end{bmatrix} \begin{matrix} H_1 \\ H_2 \\ H_3 \\ H_4 \\ H_5 \\ H_6 \end{matrix}$$

1

ReLU

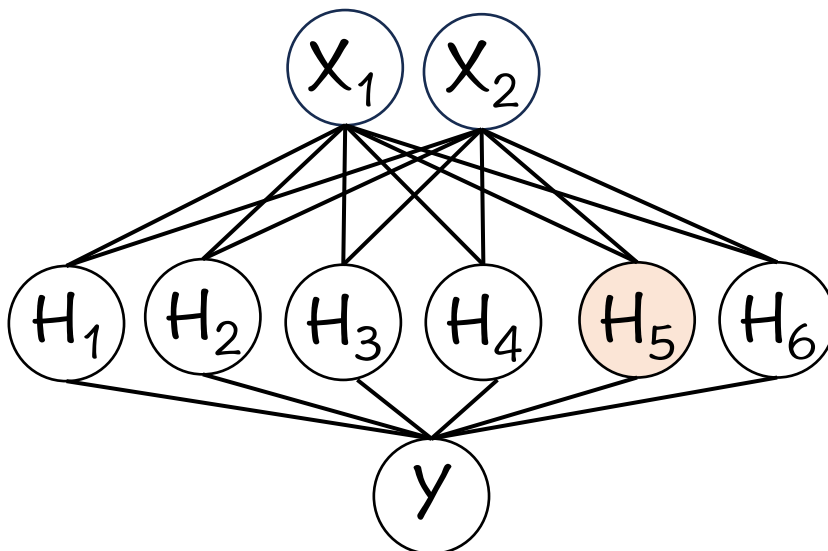
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 8 \end{bmatrix}$$

$\approx$

$$\begin{bmatrix} 8 \end{bmatrix}$$

y



$$\begin{bmatrix} -1 & 0 \end{bmatrix}$$

Hidden Layer

Exercise 25

$$\begin{matrix} 3 \\ 2 \end{matrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$$

1

0	1	0
1	0	0
-1	1	0
0	0	2
0	-1	0
1	-1	0

2
3
-1
2
-2
1

ReLU
 $\approx$

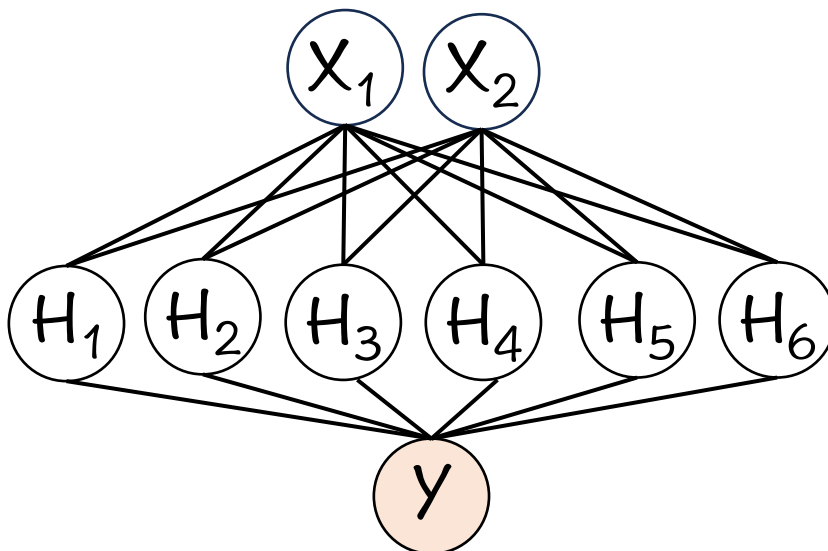
2	H_1
3	H_2
0	H_3
2	H_4
0	H_5
1	H_6

1

ReLU
 $\approx$

y

1	1	1	1	1	1	0
---	---	---	---	---	---	---



8	8
---	---